

Optimization

2.2 Golden section search

José Firmino Aguilar Madeira

Where to place the two points?

In 2.1: f unimodal on $[1, 4]$, $x_1 = 2$, $x_2 = 3$; since $f(2) > f(3)$, we kept $[2, 4]$.

The point $x_2 = 3$ survived, with f already computed — but it lies in the **middle** of $[2, 4]$. If we split into thirds again (2.67 and 3.33), we would not reuse it: **two new evaluations** per iteration, to reduce the interval to $2/3$.

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We have two interior points and want to reduce the interval many times, **reusing** one of the evaluations. Where should x_1 and x_2 be?

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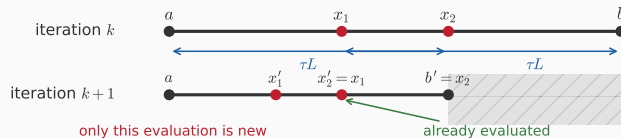
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Two requirements:

1. **Symmetry**: $x_1 = b - \tau L$, $x_2 = a + \tau L$ ($L = b - a$). Whatever the comparison, the new interval has length τL .
2. **Reuse**: the surviving point must be exactly where the next iteration needs one of its two points.

If both hold, each reduction costs **a single evaluation of f** . Which τ achieves this?

Reusing a point: $\tau^2 = 1 - \tau$



Iteration k : $x_1 = a + (1 - \tau)L$, $x_2 = a + \tau L$. Suppose $f(x_1) < f(x_2)$: we keep $[a, x_2]$, of length τL .

At iteration $k+1$ the right-hand point must be at $x'_2 = a + \tau(\tau L) = a + \tau^2 L$. For it to be the old x_1 :

$$\tau^2 L = (1 - \tau)L \iff \boxed{\tau^2 = 1 - \tau}$$

(the other case, by symmetry, gives the same condition).

$$\tau^2 + \tau - 1 = 0 \Rightarrow \tau = \frac{\sqrt{5} - 1}{2} \approx 0.618.$$

reduction factor $\tau \approx 0.618$
per **one** new evaluation

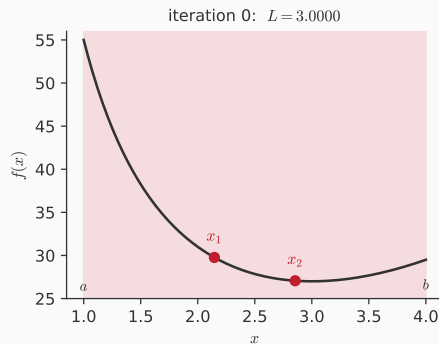
The reciprocal $1/\tau = (1 + \sqrt{5})/2 \approx 1.618$ is the *golden ratio*; hence the name of the method.

One iteration by hand

$f(x) = x^2 + 54/x$ on $[1, 4]$, $L_0 = 3$, $\tau \approx 0.618$:

$$x_1 = 4 - \tau \cdot 3 = 2.146, \quad x_2 = 1 + \tau \cdot 3 = 2.854,$$

$$f(x_1) = 29.77, \quad f(x_2) = 27.07.$$



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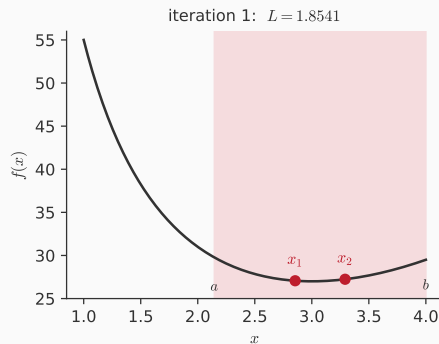
$$x_1 = 4 - \tau \cdot 3 = 2.146, \quad x_2 = 1 + \tau \cdot 3 = 2.854,$$

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Since $f(x_1) > f(x_2)$: new interval $[2.146, 4]$,
 $L_1 = 1.854 = \tau L_0$.

The new left-hand point would be $4 - \tau \cdot 1.854 = 2.854$:
it is the old x_2 , already evaluated. We only need

$$x'_2 = 2.146 + \tau \cdot 1.854 = 3.292, \quad f(x'_2) = 27.24.$$



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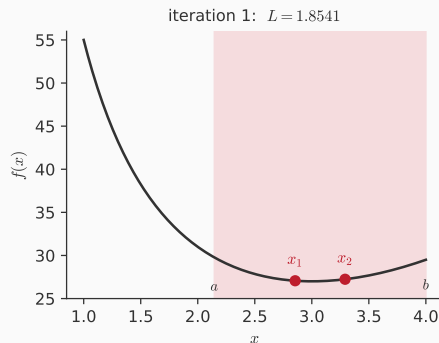
$$f(x_1) = 29.77, \quad f(x_2) = 27.07.$$

Since $f(x_1) > f(x_2)$: new interval $[2.146, 4]$,
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$$x'_2 = 2.146 + \tau \cdot 1.854 = 3.292, \quad f(x'_2) = 27.24.$$

Thirds (2.1): factor $2/3$ with 2 new evaluations.
Golden section: factor 0.618 with 1 new evaluation.



The algorithm

Golden section on $[a, b]$, tolerance ε_x

```
1:  $\tau \leftarrow (\sqrt{5} - 1)/2$ ;  $L \leftarrow b - a$ 
2:  $x_1 \leftarrow b - \tau L$ ;  $x_2 \leftarrow a + \tau L$   $\triangleright x_1 < x_2$ 
3:  $f_1 \leftarrow f(x_1)$ ;  $f_2 \leftarrow f(x_2)$ 
4: while  $L > \varepsilon_x$  do
5:   if  $f_1 < f_2$  then  $\triangleright$  minimum in  $[a, x_2]$ 
6:      $b \leftarrow x_2$ ;  $x_2 \leftarrow x_1$ ;  $f_2 \leftarrow f_1$ 
7:      $L \leftarrow b - a$ ;  $x_1 \leftarrow b - \tau L$ ;  $f_1 \leftarrow f(x_1)$ 
8:   else  $\triangleright f_1 \geq f_2$ : minimum in  $[x_1, b]$ 
9:      $a \leftarrow x_1$ ;  $x_1 \leftarrow x_2$ ;  $f_1 \leftarrow f_2$ 
10:     $L \leftarrow b - a$ ;  $x_2 \leftarrow a + \tau L$ ;  $f_2 \leftarrow f(x_2)$ 
11:   end if
12: end while
13: return  $x^* \approx (a + b)/2$ 
```

- Lines 6 and 9: the surviving point **changes role** ($x_1 \rightarrow x_2$ or $x_2 \rightarrow x_1$) and carries its value of f : two evaluations to start, then **one per reduction**.
- **Tie** $f_1 = f_2$: goes to the *else* branch. This is correct: in that case $x^* \in [x_1, x_2] \subset [x_1, b]$ (2.1). No third branch is needed.
- **Without unimodality the guarantee is lost**: the comparisons may eliminate the region of the desired minimum.

How many reductions?

Let $L_0 = b - a$ be the initial interval. After k reductions, $L_k = \tau^k L_0$ ($\tau \approx 0.618$) and, taking the midpoint $m_k = (a_k + b_k)/2$ as the estimate, $|m_k - x^*| \leq L_k/2$. With ε_x the tolerance on the **interval length** (as in the **while** of the algorithm), to guarantee $L_N \leq \varepsilon_x$ — and hence $|m_N - x^*| \leq \varepsilon_x/2$ — it suffices that

$$N \geq \frac{\ln(\varepsilon_x/L_0)}{\ln \tau} = \frac{\ln(L_0/\varepsilon_x)}{0.4812}.$$

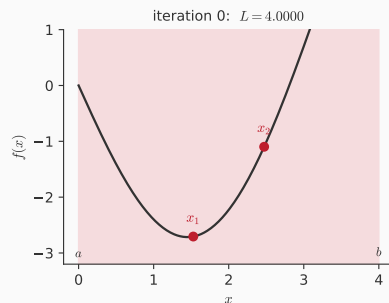
Example

$L_0 = 1$, $\varepsilon_x = 0.01$: $N \geq \ln 100/0.4812 = 9.57$, so $N = 10$ reductions ($0.618^{10} = 0.0081$), with $2 + 10 = 12$ evaluations of f .

- **Linear** convergence with ratio 0.618: each extra decimal digit costs ≈ 4.8 iterations.
- N *does not depend on* f — only on L_0 and ε_x : predictable, but slow compared with Newton (2.5). Bisection on f' reduces by 0.5 per iteration, but needs f' .

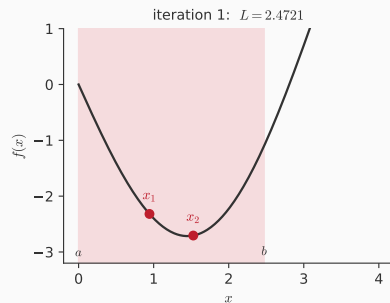
Example 1: $f(x) = -3 \sin x + x^2/8$ on $[0, 4]$

k	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$	L_k
0	0.0000	4.0000	1.5279	2.4721	-2.7054	-1.0977	4.0000



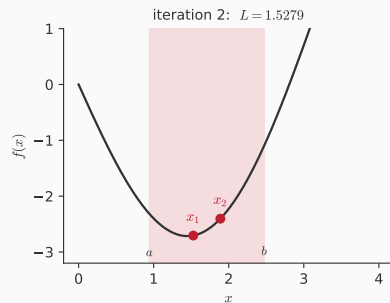
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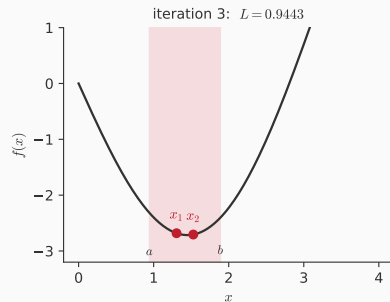
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2	0.9443	2.4721	1.5279	1.8885	-2.7054	-2.4040	1.5279



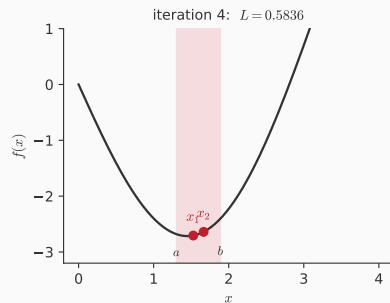
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3	0.9443	1.8885	1.3050	1.5279	-2.6818	-2.7054	0.9443



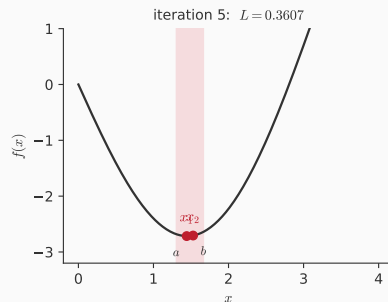
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4	1.3050	1.8885	1.5279	1.6656	-2.7054	-2.6397	0.5836



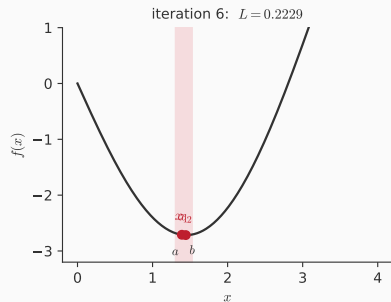
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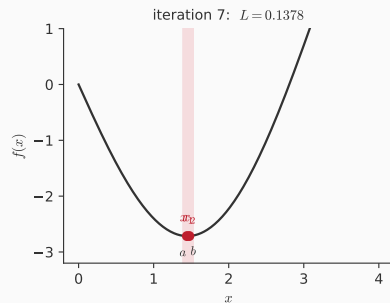
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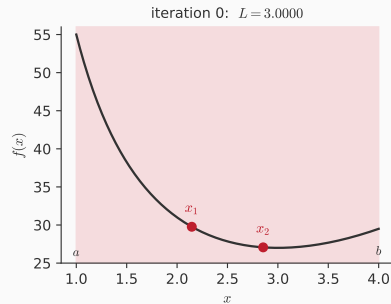
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7	1.3901	1.5279	1.4427	1.4752	-2.7152	-2.7143	0.1378

Row k has $L_k = \tau^k L_0$. After 15 reductions: $L_{15} = 4\tau^{15} = 0.0029$, $x^* \approx 1.4489$, $f(x^*) \approx -2.7153$.



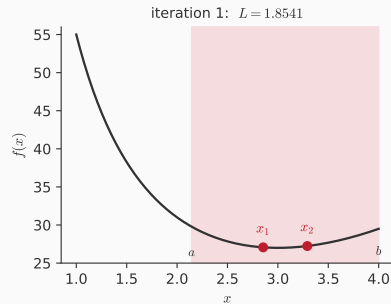
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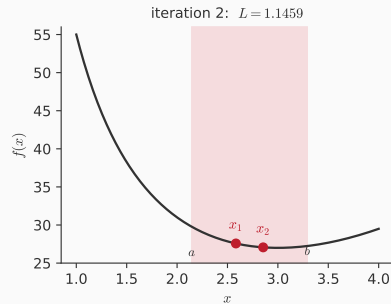
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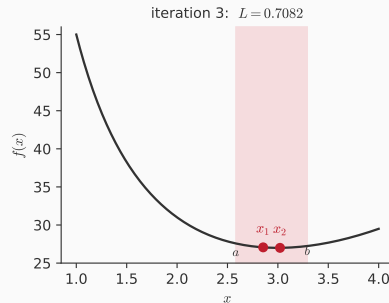
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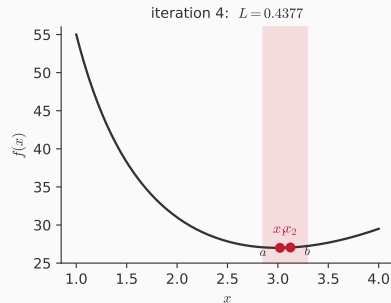
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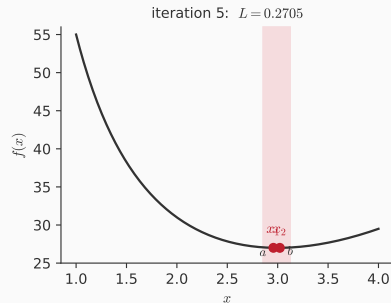
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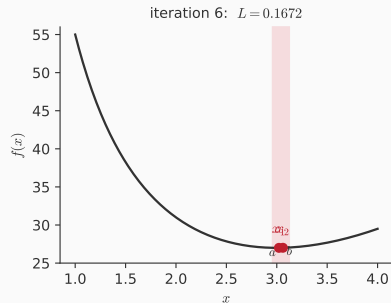
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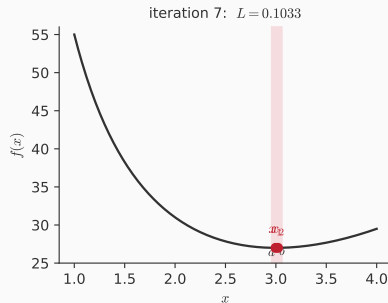
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After 15 reductions: $L_{15} = 3\tau^{15} = 0.0022$, $x^* \approx 3.0002$,

$f(x^*) \approx 27.0000$. Exact solution:

$$f'(x) = 2x - 54/x^2 = 0 \Rightarrow x^* = 3, f(3) = 27.$$



On the computer

The core, in MATLAB (N reductions):

```
tau = (sqrt(5)-1)/2;  L = b - a;
x1 = b - tau*L;  f1 = f(x1);    % x1 < x2
x2 = a + tau*L;  f2 = f(x2);
for k = 1:N
    if f1 < f2                % minimum in [a,x2]
        b = x2;  x2 = x1;  f2 = f1;
        L = b - a;  x1 = b - tau*L;  f1 = f(x1);
    else                    % minimum in [x1,b]
        a = x1;  x1 = x2;  f1 = f2;
        L = b - a;  x2 = a + tau*L;  f2 = f(x2);
    end
end
x = (a + b)/2;
```

The full function (GoldenSection, with info.nfev and the table printout) is in the course companion code.

One evaluation per reduction (the other value is reused): $n_f = N + 2$ to locate x^* ; if we also evaluate f at the final midpoint, $n_f = N + 3$. So, for $N = 15$, $n_f = 18$.

MATLAB's `fminbnd` combines golden section with parabolic interpolation (2.3).

THE SAME IN PYTHON

```
from scipy.optimize import minimize_scalar
r = minimize_scalar(f, bounds=(0, 4),
                   method='bounded')
r.x, r.fun, r.nfev
```

`bounded` and `fminbnd` use Brent-type strategies; the details may differ. `r.nfev` counts the evaluations.

KEY TAKEAWAYS — 2.2

1. With f unimodal, two interior points allow us to **eliminate** part of the interval; only the *order* of the values of f is used.
2. Points symmetric at τL from the endpoints, with $\tau^2 = 1 - \tau$, $\tau = (\sqrt{5} - 1)/2 \approx 0.618$: the surviving point is reused, and each reduction costs **one** evaluation of f .
3. $L_k = \tau^k L_0$ and, with the midpoint m_k , $|m_k - x^*| \leq L_k/2$: linear convergence, with $N \geq \ln(L_0/\varepsilon_x)/0.481$ reductions, **independent of f** .
4. Robust (needs only unimodality), but slow: each decimal digit costs ≈ 4.8 reductions.

Next (2.3): golden section uses only the *order* of $f(x_1)$ and $f(x_2)$. What if we also use their **magnitude** — fitting a parabola through three points and jumping to its minimum? — successive quadratic interpolation (Powell's method).

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 3.

Resources

- Companion code: `GoldenSection (info.nfev)`. MATLAB: `fminbnd`.
- Python: `scipy.optimize.minimize_scalar(f, bounds=(a,b), method='bounded')` (`r.nfev`).